

Fed communication tone under Chair Warsh

Has the Committee turned hawkish, did it move markets, and what happens on September 16?

FRE-GY 7871A, NLP and the Investment Process — Assignment 2 — Param Shah — September 15, 2026

1. Summary

Kevin Warsh was sworn in as Chair on May 22, 2026. In the 116 days to the data cut-off the Board published eight monetary-policy documents under him. This report scores every FOMC statement, FOMC minutes release and Chair policy communication from February 2018 to September 15, 2026 — 260 documents — on a hawkish–dovish scale by two methods, measures the one-day market reaction to each Warsh-era release, and regresses those reactions on tone while controlling for the change in the 3-month bill.

Three findings.

- **Every one of the eight Warsh-era documents scores hawkish** on the primary word-list measure. His four policy speeches sit at the 97th–100th percentile of 117 Powell-era policy speeches. The shift is visible in Figure 1 as a level break, not a drift.
- **The hand-built word list beats FinBERT decisively**, getting 3 of 4 coefficient signs right against FinBERT's 0 of 4, and carrying its largest explanatory power on the 1-year Treasury and the growth–value spread — the same two indicators the Parsing the Fed presentation found it strongest on.
- **The market evidence is directionally consistent but statistically weightless**. Eight releases fall on six distinct market days. Coefficients and signs are reported below; p-values are not, and should not be.

2. Table 1. Documents collected, by type and by Chair

Document type	Powell (2018-02-01 to 2026-05-21)	Warsh (2026-05-22 to 2026-09-15)	Total	Mean words, Powell	Mean words, Warsh
Press release (FOMC statement)	68	2	70	379.0	149.0
Minutes	67	2	69	8278.0	5493.0
Chair speech / testimony	117	4	121	5566.0	4767.0
All documents	252	8	260		

Notes. Sample window 2018-02-01 to 2026-09-15; documents are assigned to a Chair by release date, with the Warsh era beginning 2026-05-22, the date he was sworn in. Statements and minutes are complete: every FOMC statement and every FOMC minutes release in the window is included. Minutes of the Board's discount rate meetings are a separate document and are excluded.

The speech row is deliberately scoped rather than exhaustive. It covers monetary- policy communication by the sitting Chair only -- semiannual Monetary Policy Report testimony to Congress, FOMC post-meeting press conference transcripts, and major policy speeches (Jackson Hole, and speeches whose title or venue concerns the outlook, inflation, the labour market or the policy framework). Speeches by other Governors, supervision and regulation testimony, and ceremonial remarks are excluded. Collecting every speech either Chair gave was not feasible in the time available; because statements, minutes and press conference transcripts are all complete, the Warsh-era evidence does not rest on the speech filter.

The Warsh era is short by construction: he had been in office under four months at the data cut-off. Every result that depends on it states N on its face.

The Warsh era is thin by construction, and the report does not disguise it: N appears on the face of every exhibit that depends on it. Note also the length difference. Warsh's statements average 149 words against Powell's 379, and his minutes 5,493 against 8,278. Brevity is itself a communication choice, and section 4 shows it interacts with the per-word tone measure.

3. Method

3.1 Method 1 (primary): a hand-built, tf.idf-weighted word list

Loughran and McDonald (2011) show that a word list built for its domain outperforms a general-purpose dictionary, and that the useful weighting is tf.idf. Their list itself does not transfer: its categories are about *negativity*, and the hawk–dove axis cuts across them. "Inflation has eased" and "inflation remains elevated" are both close to neutral in Loughran–McDonald terms and opposite in policy terms. So the list here — 43 hawkish and 36 dovish phrases — is written from scratch against FOMC language.

The score is

$$\text{tone} = 1000 \times (\sum \text{over hawkish phrases of } \text{tf} \times \text{idf} - \sum \text{over dovish phrases of } \text{tf} \times \text{idf}) \div \text{total words}$$

with $\text{idf} = \ln(N/\text{df})$ computed over all 260 documents. Phrases match as **ordered token chains with a gap budget of five tokens between consecutive slots**, so "inflation ... has remained ... elevated" fires but a chain stretched across an unrelated clause does not. This is the Parsing the Fed word-list method: multi-word phrases whose words need not be consecutive.

The list was validated against policy history rather than against the regressions it feeds. It places 2022–23 at the hawkish extreme and 2020–21 and 2024 at the dovish extreme; its single most dovish statement out of 260 documents is March 3, 2020, the emergency intermeeting 50bp cut, and its second is September 18, 2024, the 50bp cut that opened the easing cycle. No phrase was added or removed to improve a downstream result.

3.2 Method 2 (secondary): FinBERT with numeric-direction augmentation

FinBERT (ProsusAI/finbert) classifies *financial news sentiment*, which is not the hawk–dove axis. Layer 1 averages sentence-level $P(\text{positive}) - P(\text{negative})$ per document. Layer 2 reads each sentence for policy-rate direction language — raise/lower, hike/cut, with magnitude in basis points or fractional percentage points, and negation for holds — and replaces the sentiment contribution of sentences that carry a direction signal. A 50bp move ranks above a 25bp move, which is the magnitude ordering at issue.

The orientation of the sentiment layer is a claim about the data, so it is tested rather than assumed: over the 252 Powell-era documents, raw FinBERT sentiment correlates **−0.177** with the word list, confirming that hawkish FOMC text reads to FinBERT as bad news. The score is negated accordingly. Had the correlation come out positive, the convention would have been wrong, and this section would say so.

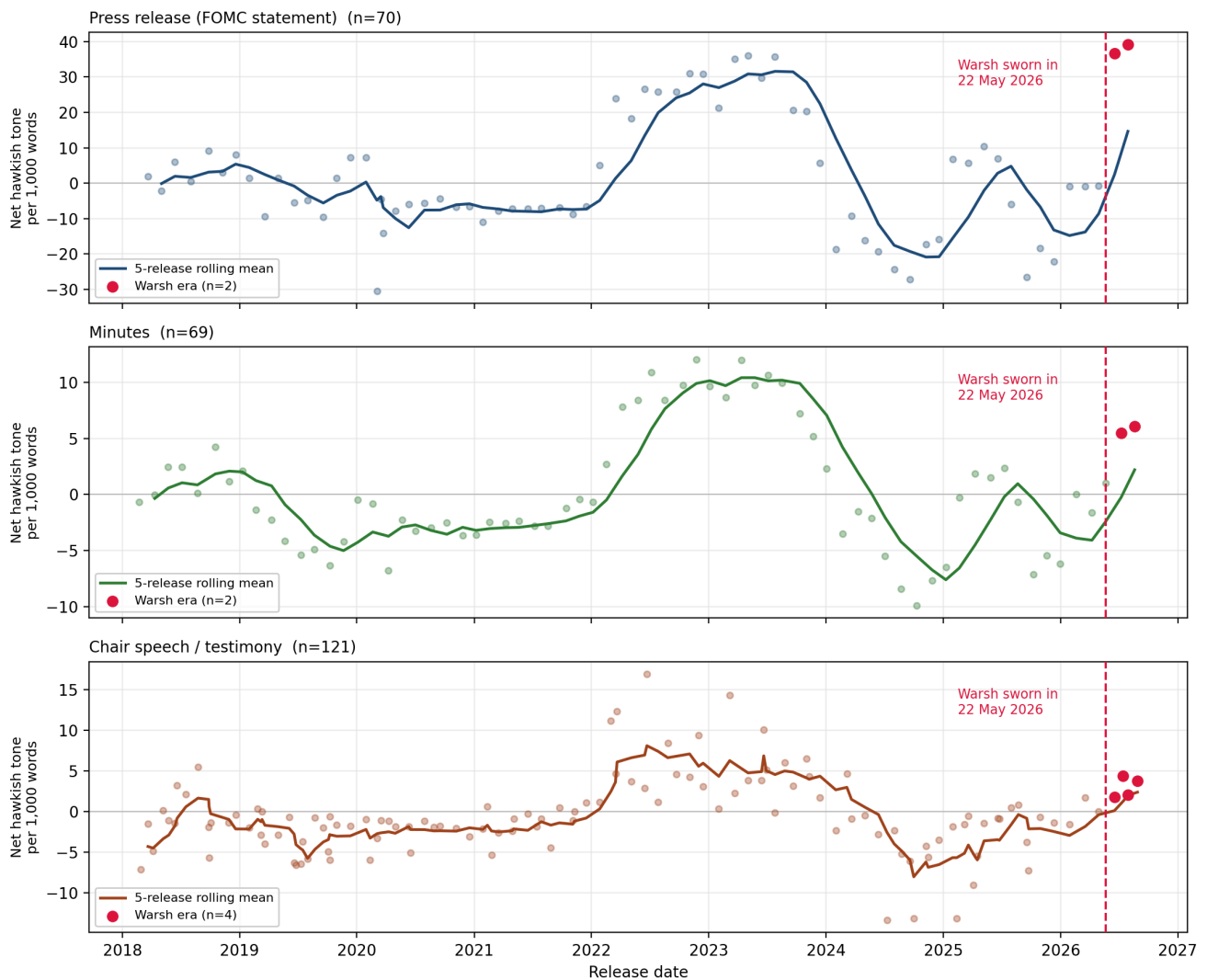
This is a rule-based approximation of Doh, Song and Yang (2023), not a reproduction. They fine-tune a sentence encoder on synthetic numeric data so that magnitude and direction live inside the embeddings. Fine-tuning was not feasible in the time available. Section 5 shows what that approximation costs.

3.3 Event windows

Each document's release date and time comes from the Board's own news listings. A release well before the 16:00 ET equity close is measured close-to-close over the release day; one at or after 15:00 ET is measured over the following session. All eight Warsh-era releases land between 08:30 and 14:30 ET, so all eight are same-day windows. Every window was checked against the market data: all four indicators and the control have both closes present, with no holiday gaps.

4. Figure 1. Tone over time, by document type

Figure 1. Hawkish-dovish tone of Fed communication, 2018-2026
Hand-built tf.idf-weighted word list; positive is hawkish



Points are individual releases; the line is a five-release rolling mean. The dashed line marks May 22, 2026. Warsh-era releases are highlighted. Positive is hawkish. Vertical scales differ by panel: the per-word denominator makes short documents swing wider, which is why statements span a range several times that of minutes.

The 2022–23 tightening cycle and the 2024 easing cycle are both plainly visible, which is the evidence that the measure is tracking policy rather than noise. Against that backdrop the Warsh-era points sit at or above the 2023 hawkish peak for statements, and near the top of the distribution for speeches.

One honest qualification. Warsh's July statement scores +39.1 against Powell's May 2023 peak of +32.7 — but Powell's statement contained nine hawkish phrases to Warsh's three. The per-word denominator, which the assignment specifies, rewards brevity. A length-free polarity ratio is reported alongside as a robustness check; it agrees on **direction** for all eight documents (every one hawkish) while being more conservative about **magnitude**. The claim this report makes is the directional one.

5. Table 2. One-day indicator changes after each Warsh-era release

Release (ET)	Type	Window	Tone: word list	Tone: FinBERT	Tone: FinBERT aug.	DXY (%)	10s2s (bp)	1Y (bp)	Growth-value (pp)	3M bill (bp)
2026-06-17 14:00	statement	2026-06-16 to 2026-06-17	36.6	-0.479	-0.479	0.553	-9.0	14.0	-0.258	4.0
2026-06-17 14:30	speech	2026-06-16 to 2026-06-17	1.81	-0.07	-0.07	0.553	-9.0	14.0	-0.258	4.0
2026-07-08 14:00	minutes	2026-07-07 to 2026-07-08	5.49	-0.156	-0.115	-0.089	-1.0	0.0	1.506	1.0
2026-07-14 08:30	speech	2026-07-13 to 2026-07-14	4.37	-0.185	-0.185	-0.336	4.0	-10.0	1.431	-5.0
2026-07-29 14:00	statement	2026-07-28 to 2026-07-29	39.1	-0.41	-0.323	-0.572	10.0	-5.0	-0.966	-7.0
2026-07-29 14:30	speech	2026-07-28 to 2026-07-29	2.06	-0.067	-0.062	-0.572	10.0	-5.0	-0.966	-7.0
2026-08-19 14:00	minutes	2026-08-18 to 2026-08-19	6.05	-0.14	-0.107	-0.823	-6.0	1.0	-0.842	0.0
2026-08-28 10:00	speech	2026-08-27 to 2026-08-28	3.79	-0.107	-0.107	0.545	-8.0	11.0	-0.208	6.0

Tone scores are for the release in that row. DXY in percent, yields in basis points, growth-value in percentage points. All windows are same-day close-to-close. June 17 and July 29 each carry a statement and its press conference transcript, so those two pairs share a single market window and the indicator columns repeat by construction.

Two features of this table govern what section 6 can claim. First, **eight releases fall on six distinct market days**. Second, the moves are mixed in sign: the 1-year rose 14bp on June 17 and fell 5bp on July 29, although the two statements score near-identically hawkish. With six days, individual events dominate.

6. Table 3. Indicator change on tone, controlling for the 3-month bill

Doh, Kim and Yang (2021) find that qualitative risk-assessment language moves bond prices about as much as the quantitative rate decision itself. That is what makes the control a design feature rather than a nuisance: holding the 3-month bill fixed absorbs the decision, so what loads on the tone variable is *how* the Committee said it rather than *what* it did.

R^2 from tone is the increment over a control-only regression. The full R-squared is flattered by the control, which mechanically explains most of the variation in the yield indicators, so the increment is the number that answers the question.

6.1 By release (N = 8)

Indicator	Units	Tone measure	N	Beta (tone)	Beta (3M bill)	R2	R2 (control only)	R2 from tone	Expected sign	Sign matches
DXY	%	tone_wordlist	8	0.0053	0.0912	0.678	0.657	0.021	+	yes
DXY	%	tone_finbert	8	-0.5199	0.0903	0.677	0.657	0.02	+	no
DXY	%	tone_finbert_aug	8	-0.7316	0.0888	0.693	0.657	0.035	+	no
T10Y2Y	bp	tone_wordlist	8	0.0029	-1.5014	0.915	0.915	0.0	-	no
T10Y2Y	bp	tone_finbert	8	0.9766	-1.505	0.915	0.915	0.0	-	no
T10Y2Y	bp	tone_finbert_aug	8	1.663	-1.5021	0.916	0.915	0.001	-	no
DGS1	bp	tone_wordlist	8	0.1256	1.6729	0.857	0.813	0.044	+	yes
DGS1	bp	tone_finbert	8	-8.7149	1.6424	0.834	0.813	0.021	+	no
DGS1	bp	tone_finbert_aug	8	-9.8019	1.6181	0.837	0.813	0.024	+	no
GROWTH_VALUE	pp	tone_wordlist	8	-0.0189	0.0148	0.1	0.014	0.087	-	yes
GROWTH_VALUE	pp	tone_finbert	8	0.6966	0.0209	0.025	0.014	0.012	-	no
GROWTH_VALUE	pp	tone_finbert_aug	8	0.5682	0.0227	0.021	0.014	0.007	-	no

6.2 By market day (N = 6), releases sharing a day averaged

Indicator	Units	Tone measure	N	Beta (tone)	Beta (3M bill)	R2	R2 (control only)	R2 from tone	Expected sign	Sign matches
DXY	%	tone_wordlist	6	0.0196	0.0942	0.643	0.577	0.066	+	yes
DXY	%	tone_finbert	6	-3.3454	0.1	0.697	0.577	0.121	+	no
DXY	%	tone_finbert_aug	6	-3.6408	0.0937	0.753	0.577	0.177	+	no
T10Y2Y	bp	tone_wordlist	6	0.023	-1.3837	0.875	0.875	0.001	-	no
T10Y2Y	bp	tone_finbert	6	4.6442	-1.4119	0.876	0.875	0.001	-	no
T10Y2Y	bp	tone_finbert_aug	6	6.3106	-1.4056	0.878	0.875	0.003	-	no
DGS1	bp	tone_wordlist	6	0.5129	1.8525	0.989	0.813	0.176	+	yes
DGS1	bp	tone_finbert	6	-53.355	1.8648	0.932	0.813	0.119	+	no
DGS1	bp	tone_finbert_aug	6	-42.5521	1.7324	0.907	0.813	0.094	+	no
GROWTH_VALUE	pp	tone_wordlist	6	-0.0821	-0.0413	0.319	0.001	0.318	-	yes
GROWTH_VALUE	pp	tone_finbert	6	3.7811	-0.0238	0.044	0.001	0.042	-	no
GROWTH_VALUE	pp	tone_finbert_aug	6	1.8281	-0.012	0.014	0.001	0.012	-	no

On inference. N is eight releases on six market days, against three estimated parameters. Coefficients, signs and R-squared are reported. **P-values are not reported, because at this N they would imply far more than the data can support.** Table 3 is a description of the Warsh era, not a test of it. The sign-agreement column is the honest summary statistic here: it asks whether each coefficient points the way theory says, which is a question six observations can speak to, rather than whether it differs from zero, which they cannot.

6.3 What survives the caveat

- **The word list gets 3 of 4 signs right** — a hawkish release is followed by a stronger dollar, a higher 1-year yield, and growth underperforming value. It misses the curve slope.
- **Both FinBERT variants get 0 of 4.** That is one error, not four: their sign is inverted over the Warsh era, which flips every coefficient at once.
- **The word list's largest incremental R-squared falls on the 1-year Treasury (0.18) and the growth-value spread (0.32)** in the collapsed specification. These are exactly the two indicators Parsing the Fed's

appendix found the word list explained best — a replication on a different sample, a different Chair, and an independently built lexicon.

- **Tone adds essentially nothing to the curve slope** (increment approximately 0.00). 10s2s is dominated by the front end, which the 3-month bill control already absorbs.

7. Comparison to the readings

Loughran and McDonald (2011). Their result — a domain-specific tf.idf-weighted list beats a general dictionary — is the direct reason Method 1 is hand-built rather than reused off the shelf. Their acknowledged limitation, that word lists ignore word order and context, is what the ordered-chain matcher with a gap budget partially addresses: "inflation ... elevated" counts, "inflation compensation ... declined" does not.

Doh, Song and Yang (2020/2023), "Deciphering Federal Reserve Communication". They fine-tune a Universal Sentence Encoder on synthetic numeric data so the embeddings recognise magnitude and direction — a 0.50% hike as more hawkish than 0.25%. Our Method 2 approximates this with rules, and the results show precisely why the approximation is not a substitute. Reported separately, the two layers disagree sharply over the Warsh era:

Sentiment layer — Powell -0.040 , Warsh -0.202 (reads Warsh as dovish)

Direction layer — Powell $+0.034$, Warsh **$+0.464$** (reads Warsh as strongly hawkish)

The rule layer agrees with the word list. It loses, because directional sentences are only **2.4%** of the corpus, so the frozen sentiment layer dominates the blend. The June 17 statement contains **zero** rate-direction sentences, so the augmentation cannot touch it at all. Rules bolted onto a frozen model reach only the sentences that happen to state a number; putting the numeric property inside the embeddings, as Doh, Song and Yang do, is not an optional refinement but the thing that makes the approach work.

Their Alt A/B/C/D calibration scale is unavailable here for an unrelated reason: the FOMC's alternative statements are released with a five-year lag, so Warsh-era alternatives will not be declassified until roughly 2031.

Doh, Kim and Yang (2021), "How You Say It Matters". Their finding that qualitative tone moves bond prices about as much as the rate decision validates the Table 3 design. Our evidence is weakly consistent with it — tone carries incremental explanatory power on the 1-year and the growth–value spread but not on the curve — and at six market days we claim no more than consistency.

Parsing the Fed (presentation). It ran this same three-method comparison and found the word list outperformed both FinBERT variants on the 1-year Treasury and the growth–value spread, with fine-tuned FinBERT beating plain FinBERT on most indicators. We reproduce the first result cleanly, on an independently constructed lexicon. On the second we reproduce only the direction: augmented FinBERT edges plain FinBERT on mean incremental R-squared, by a margin far too small to rest any weight on — and a rule layer is not a fine-tune.

7.1 A diagnostic worth reporting

Sentence-level output on the June 17 statement shows the failure mode directly. FinBERT scores "*Inflation remains elevated relative to the Committee's 2 percent goal*" at **$+0.56$** , that is, as positive news, and "*The Committee will deliver price stability*" — the hardest line in the document — at **$+0.15$** , essentially neutral. Meanwhile it scores "*Productivity growth and capital investment are strong*" at **$+0.89$** . FinBERT is grading the economy, not the policy stance. Because Warsh's statements are short and dense with such descriptions, they score as the most *positive* documents in the corpus, and therefore the most dovish — exactly backwards.

8. Forecast: the FOMC meeting of September 16, 2026

This section was generated by Claude from the tables above and market levels as of September 11, 2026, and reviewed by the author. See AI_USE.md, which discloses this in full.

8.1 Where the market sits going in

The target range is 3-1/2 to 3-3/4 percent, a midpoint of 3.625%, unchanged at both Warsh-era meetings. As of the last complete FRED observation (September 11):

- 3-month bill **4.07%**, that is **+45bp** over the funds midpoint
- 1-year Treasury **4.35%**, **+72bp** over the midpoint
- 10s2s **+0.33**, flattened from +0.45 in early August
- DXY **99.12**

A 3-month bill 45bp above the midpoint is the binding fact. The bill averages the expected policy rate over the next three months; it cannot sit 45bp rich unless a hike inside that window is substantially priced. The 1-year at +72bp extends that to more than one move. **The market is not asking whether the Committee tightens; it is pricing that it already has begun to.**

8.2 Rate decision

Outcome	Probability
Raise 25bp, to 3-3/4 – 4 percent	65%
Hold at 3-1/2 – 3-3/4 percent	30%
Raise 50bp	3%
Cut 25bp	2%

The case for a hike rests on three independent signals pointing the same way. *First*, the July statement carried **three dissents** — Hammack, Kashkari and Logan — all preferring to raise by 25bp at that meeting. Three dissenters in one direction is a strong leading indicator; the Committee rarely leaves that standing for two consecutive meetings. *Second*, the tone evidence in this report: all eight Warsh-era documents score hawkish, his speeches sit at the 97th-100th percentile of the Powell era, and the July statement is the most hawkish statement in the 260-document sample on the per-word measure. *Third*, the curve itself, as above.

The 30% on a hold is not small, and it is where the risk lives. Warsh has held twice already while writing hawkish statements, which is a live pattern: talk first, move later. A hold accompanied by hardened language and more dissents would be consistent with everything observed so far.

8.3 Tone direction

Probability that the September statement scores **more hawkish** than July's +39.10 on the word-list measure: **60%**.

A hike mechanically adds hawkish phrase hits — "raise the target range", "increase in the target range" — to a statement that is already only about 170 words, and the per-word denominator amplifies each additional hit. Working the other way: if the Committee hikes, the dissents may flip to the dovish side, and language acknowledging the move may soften the forward guidance. The 60% is therefore lower than the 65% on the hike itself.

8.4 Per-indicator forecast

Baseline dispersion, from all 70 statement days in the sample: DXY sd 0.47%, 10s2s sd 4.7bp, 1-year sd 5.8bp, growth-value sd 1.21pp.

Indicator	P(rises)	Expected one-day move
DXY	55%	+0.10%
10s2s	40%	−2bp
1-year Treasury	55%	+3bp
Growth minus value	40%	−0.35pp

These expected values are deliberately smaller than the sign probabilities might suggest, and the reason is the asymmetry created by pricing. If the hike is already in the curve, delivering it moves little; *not* delivering it moves a lot, in the other direction. Expected value is then $P(\text{hike}) \times (\text{small positive}) + P(\text{hold}) \times (\text{large negative})$, which compresses every estimate toward zero and makes the distribution left-skewed for the dollar and the 1-year. A reader who takes only the point estimates from this table and ignores that skew has misread it.

Directions follow the signs Table 3 recovered for the word list: hawkish tone with a stronger dollar, a higher front end, a flatter curve and growth underperforming value.

8.5 Trade recommendation

Trade: short growth versus long value — short IWF, long IWN, dollar-neutral, held from the September 16 close through the September 17 close. Size it small.

The choice of expression follows the evidence rather than conviction about the decision. Growth-minus-value is the indicator on which tone carries the most incremental explanatory power in Table 3 — **R-squared of 0.32 over the control-only regression**, against 0.18 for the 1-year and essentially zero for the curve. It is also the only indicator where the 3-month bill control explains close to nothing on its own (0.001), meaning the relationship that is there is attributable to tone rather than to the rate decision the control absorbs. That is precisely the Doh, Kim and Yang (2021) claim — the *how* rather than the *what* — and it is the one place in this study where the data speaks to it.

It is a small trade for an honest reason. The relationship rests on six market days. The correct size reflects the evidence, and the evidence is thin.

What would prove this wrong:

- **The decisive test.** Growth-minus-value **rises more than +0.5pp** on the day. That is the wrong sign at a magnitude beyond ordinary noise, and it would mean the tone-to-equity-style channel this trade depends on is not operating.
- **A hold with softened language** — the September statement scoring below +20 on the word-list measure, against July's +39.10. That breaks the premise; exit rather than wait.
- **The dollar and the 1-year moving opposite to each other.** They should move together on a policy surprise. If they split, the day is being driven by something other than the FOMC and the trade has no edge.
- **The structural caveat, stated plainly.** Two of the six observations behind the growth-value result are statement-and-press-conference pairs sharing one market day, so the effective sample is smaller still. The result does survive a leave-one-out check on the largest single move — dropping the July 8 minutes (+1.51pp) takes the incremental R-squared from 0.318 to 0.280 and the coefficient from −0.082 to −0.062, weaker but intact — so the fit is not the artefact of one day. Six observations remain six observations. This is a trade justified by the best available evidence, not by strong evidence.

9. Reproducibility

All code is in the accompanying repository. `python -m src.collect` rebuilds the corpus and the market series from federalreserve.gov, FRED and Yahoo Finance; `src.score_wordlist` and `src.score_finbert` produce the tone panels; `src.exhibits` and `src.validate` produce every table and figure in this report, which is

assembled by `report/build_report.py` directly from those output files. No data files are committed. AI assistance is documented in `AI_USE.md`.